Kshitiz Gupta

Java Assignment

Basic Java

Question 1

Question:
Write a program to calculate the area of a circle, rectangle, or triangle based on user input.

```
package Kshitiz_Gupta_java_training.session1.basics;


import java.util.Scanner;


public class AreaCalculator {


    public double calculateCircleArea(double radius) {

        return Math.PI * radius * radius;

    }


    public double calculateRectangleArea(double length, double
width) {

        return length * width;

    }


    public double calculateTriangleArea(double base, double height)
{

        return 0.5 * base * height;
```

```
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        AreaCalculator ac = new AreaCalculator();

        System.out.println("Choose shape: 1.Circle 2.Rectangle
3.Triangle");
        int choice = sc.nextInt();

        switch (choice) {

            case 1:
                System.out.println("Enter radius:");
                double r = sc.nextDouble();
                System.out.println("Area: " +
ac.calculateCircleArea(r));
                break;

            case 2:
                System.out.println("Enter length and width:");
                double l = sc.nextDouble();
                double w = sc.nextDouble();
                System.out.println("Area: " +
ac.calculateRectangleArea(l, w));
```

```
                break;


            case 3:

                System.out.println("Enter base and height:");

                double b = sc.nextDouble();

                double h = sc.nextDouble();

                System.out.println("Area: " +
ac.calculateTriangleArea(b, h));

                break;


            default:

                System.out.println("Invalid choice");

        }


        sc.close();

    }

}
```

```
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>  & 'C:\Program Files\Java\jdk-20\bin\java.exe' '-
XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Kshitiz Gupta\AppData\Roaming\Code\User\workspaceSt
orage\9da1682bd86be2ea0f83b442e9227cd4\redhat.java\jdt_ws\nucleusteq_assignments_3e990c0a\bin' 'Kshitiz_Gup
ta_java_training.session1.basics.AreaCalculator'
Choose shape: 1.Circle 2.Rectangle 3.Triangle
1
Enter radius:
7
Area: 153.93804002589985
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 2

Question:

Create a program to check if a number is even or odd.

```
package Kshitiz_Gupta_java_training.session1.basics;

import java.util.*;


public class OddEven {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        if(num%2==0) {

            System.out.println("number is even");

        }

        else {

            System.out.println("number is odd");

        }

    }

}
```

```
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignme
\Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExc
\Users\Kshitiz Gupta\AppData\Roaming\Code\User\worksp
ea0f83b442e9227cd4\redhat.java\jdt_ws\nucleusteq_assi
hitiz_Gupta_java_training.session1.basics.OddEven'
Enter the number
17
number is odd
```

Question 3

Question:

Implement a program to find the factorial of a given number.

```java
package Kshitiz_Gupta_java_training.session1.basics;

import java.util.*;


public class Factorial {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num=sc.nextInt();


        int prod=1;

        for(int i=1;i<=num;i++) {

            prod=prod*i;

        }

        System.out.println("The factorial is" + " " +prod);
```

```
        sc.close();

    }

}
```

```
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>  c:; cd 'c:
itiz Gupta\Desktop\nucleusteq_assignments'; & 'C:\Program Files\Java\
\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\
pta\AppData\Roaming\Code\User\workspaceStorage\9da1682bd86be2ea0f83b4
\redhat.java\jdt_ws\nucleusteq_assignments_3e990c0a\bin' 'Kshitiz_Gup
aining.session1.basics.Factorial'
Enter the number
5
The factorial is 120
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 4

Question:

Write a program to print the Fibonacci sequence up to a specified number.

```
package Kshitiz_Gupta_java_training.session1.basics;

import java.util.*;



public class Fibbonacci {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num=sc.nextInt();

        System.out.println(fib(num));

        sc.close();
```

```
    }


    public static int fib(int n) {

        if(n==0 || n==1) {

            return n;

        }

        int val=fib(n-1)+fib(n-2);

        return val;

    }

}
```

```
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>  &
Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMes
sers\Kshitiz Gupta\AppData\Roaming\Code\User\workspaceStorage
f83b442e9227cd4\redhat.java\jdt_ws\nucleusteq_assignments_3e9
z_Gupta_java_training.session1.basics.Fibbonacci'
Enter the number
7
13
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 5

Question:

Use loops to print patterns like a triangle or square.

```
package Kshitiz_Gupta_java_training.session1.basics;

import java.util.*;
```

```
public class Loops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter 1 for triangle and 2 for square");

        int choice=sc.nextInt();



        switch(choice) {

            case 1: System.out.println("enter the number of lines");

                    int n=sc.nextInt();

                    for(int i=1;i<=n;i++) {

                        for(int j=1;j<=i;j++) {

                            System.out.print("* ");

                        }

                        System.out.println();

                    }

                    break;


            case 2: System.out.println("enter the number of lines");

                    int n2=sc.nextInt();

                    for(int i=1;i<=n2;i++) {

                        for(int j=1;j<=n2;j++) {

                            System.out.print("*");

                        }

                        System.out.println();

                    }
```

```
                    break;


            default : System.out.println("Invalid choice");

        }

    }

}
```

```
Enter 1 for triangle and 2 for square
1
enter the number of lines
5
*
* *
* * *
* * * *
* * * * *
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq assignments>
```

Data Types and Operators

Question 1

Question:
Explain the difference between primitive and reference data types with examples.

```
package Kshitiz_Gupta_java_training.session1.DataTypeAndOperators;



public class Datatype {

    public static void main(String[] args) {

        //in primitive data type they store the acutal value of the
data

        int a=10;

        int b=a;

        b=20;

        System.out.println(a);


        //in reference data type they store the address of the data

        int[] arr={1,2,3};

        int[] arr2=arr;

        arr2[0]=7;

        System.out.println(arr[0]);

    }

}
```

```
aining.session1.DataTypeAndOperators.Datatype'
10
7
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq assignments>
```

Question 2

Question:

Write a program to demonstrate the use of arithmetic, logical, and relational operators.

```java
package Kshitiz_Gupta_java_training.session1.DataTypeAndOperators;

import java.util.*;


public class Operators {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.println("Enter the first no");

        int num1=sc.nextInt();


        System.out.println("Enter the second no");

        int num2=sc.nextInt();


        //Arithmetic operators

        System.out.println("Arithmetic operations results are:-");

        System.out.println("Addition " + (num1+num2));

        System.out.println("Subtraction " + (num1-num2));

        System.out.println("Multiplication " + (num1*num2));
```

```
        System.out.println("Modulas " + (num1%num2));


        if(num2!=0) System.out.println("Division " + num1/num2);

        else System.out.println("Wrong input value for division");


        //Relational operator

        System.out.println("Relational operators results are:-");

        System.out.println(num1==num2);

        System.out.println(num1>=num2);

        System.out.println(num1<=num2);

        System.out.println(num1>num2);

        System.out.println(num1<num2);

        System.out.println(num1!=num2);


        //logical operators

        System.out.println("logical operators results are:-");

        System.out.println(num1>0 && num2>0);

        System.out.println(num1>0 || num2>0);

        System.out.println(!(num1>0));


    }

}
```

```
Enter the first no
10
Enter the second no
4
Arithmetic operations results are:-
Addition 14
Subtraction 6
Multiplication 40
Modulas 2
Division 2
Relational operators results are:-
false
true
false
true
false
true
logical operators results are:-
true
true
false
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 3

Question:

Create a program to convert a temperature from Celsius to Fahrenheit and vice versa.

```
package Kshitiz_Gupta_java_training.session1.DataTypeAndOperators;

import java.util.*;



public class TemperatureConversion {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter 1 for converthing celsius to
fahrenheit and 2 for coverting fahrenheit to celsius");

        int choice = sc.nextInt();



        switch(choice) {

            case 1 : System.out.println("Enter the temperature in
celsius");

                double cel=sc.nextInt();

                System.out.println(celToFah(cel));

                break;


            case 2 : System.out.println("Enter the temperature in
fahrenheit");

                double fah=sc.nextInt();

                System.out.println(fahToCel(fah));

                break;


            default : System.out.println("Invalid choice");

        }

    }

    public static double celToFah(double c) {

        return (9.0/5) * c + 32;

    }


    public static double fahToCel(double f) {

        return (5.0/9) * (f - 32);
```

```
    }

}
```

```
Output                                                    Clear
Enter 1 for converthing celsius to fahrenheit and 2 for coverting fahrenheit
    to celsius
2
Enter the temperature in fahrenheit
107
41.66666666666667
```

Control Flow Statements

Question 1

Question:
Write a program to check if a given number is prime using an if-else statement.

```
package Kshitiz_Gupta_java_training.session1.ControlFlow;

import java.util.*;



public class PrimeNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int n=sc.nextInt();

        boolean flag=true;
```

```
        for(int i=2;i<=Math.sqrt(i);i++) {
            if(n%i==0) {
                flag=false;
                System.out.println("Number is not prime");
            }
        }

        if(flag==true) {
            System.out.println("Number is prime");
        }
    }
}
```

```
Enter the number
13
Number is prime
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 2

Question:
Implement a program to find the largest number among three given numbers using a conditional statement.

```
package Kshitiz_Gupta_java_training.session1.ControlFlow;

import java.util.*;


public class LargestNo {

    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);


        System.out.println("Enter the first number");

        int num1=sc.nextInt();


        System.out.println("Enter the second number");

        int num2=sc.nextInt();


        System.out.println("Enter the third number");

        int num3=sc.nextInt();


        if(num1>num2 && num1>num3) {

            System.out.println("The largest number is " + num1);

        }

        else if(num2>num3) {

            System.out.println("The largest number is " + num2);
```

```
        }

        else {

            System.out.println("The largest number is " + num3);

        }

    }

}
```

```
Enter the first number
13
Enter the second number
7
Enter the third number
22
The largest number is 22
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 3

Question:
Use a for loop to print a multiplication table.

```
package Kshitiz_Gupta_java_training.session1.ControlFlow;

import java.util.*;



public class MultiplicationTable {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number of which you want to
print the table");
```

```
        int num=sc.nextInt();



        for(int i=1;i<=10;i++) {

            System.out.println(num*i);

        }

    }

}
```

```
Enter the number of which you want to print the table
5
5
10
15
20
25
30
35
40
45
50
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 4

Question:

Create a program to calculate the sum of even numbers from 1 to 10 using a while loop.

```
package Kshitiz_Gupta_java_training.session1.ControlFlow;

import java.util.*;
```

```java
public class Evensum {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int i=1;

        int sum=0;

        while(i<=10) {

            if(i%2==0) {

                sum=sum+i;

            }

            i++;

        }

        System.out.println(sum);

    }

}
```

```
30
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Arrays

Question 1

Question:
Write a program to find the average of elements in an array.

```
package Kshitiz_Gupta_java_training.session1.Arrays;

import java.util.*;


public class Average {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.println("Enter the size of array");

        int n=sc.nextInt();

        int[] arr = new int[n];


        System.out.println("Enter the values of array");

        for(int i=0;i<n;i++) {

            arr[i]=sc.nextInt();

        }


        double sum=0;

        for(int i=0;i<arr.length;i++) {

            sum=sum+arr[i];

        }
```

```
        System.out.println(sum/(double)arr.length);

    }

}
```

```
Enter the size of array
5
Enter the values of array
12 8 4 7 13
8.8
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 2

Question:

Implement a function to sort an array in ascending order using bubble sort or selection sort.

```
package Kshitiz_Gupta_java_training.session1.Arrays;

import java.util.*;


public class BubbleSortArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the size of the array");

        int size=sc.nextInt();

        int[] arr = new int[size];


        System.out.println("Enter the values in array");
```

```
        for(int i=0;i<size;i++) {
            arr[i]=sc.nextInt();
        }

        //bubble sort algorithm
        for(int i=0;i<size;i++) {
            for(int j=0;j<size-1-i;j++) {
                if(arr[j]>arr[j+1]) {
                    int temp=arr[j];
                    arr[j]=arr[j+1];
                    arr[j+1]=temp;
                }
            }
        }

        for(int i=0;i<size;i++) {
            System.out.println(arr[i]);
        }
    }
}
```

```
Enter the size of the array
5
Enter the values in array
13 9 7 3 8
3
7
8
9
13
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

```
package Kshitiz_Gupta_java_training.session1.Arrays;

import java.util.*;


public class SelectionSortArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.println("Enter the size of array");

        int size=sc.nextInt();

        int[] arr = new int[size];


        System.out.println("Enter the values of array");

        for(int i=0;i<size;i++) {

            arr[i]=sc.nextInt();

        }


        for(int i = 0; i < size - 1; i++) {
```

```
            int minIndex = i;

            for(int j = i + 1; j < size; j++) {

                if(arr[j] < arr[minIndex]) {

                    minIndex = j;

                }

            }

            int temp = arr[i];

            arr[i] = arr[minIndex];

            arr[minIndex] = temp;

        }



        for(int val:arr) {

            System.out.println(val);

        }

    }

}
```

```
Enter the size of array
5
Enter the values of array
17 7 9 15 13
7
9
13
15
17
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq assignments>
```

Question 3

Question:

Create a program to search for a specific element within an array using linear search.

```
package Kshitiz_Gupta_java_training.session1.Arrays;

import java.util.*;


public class LinearSearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.println("Enter the size of array");

        int size=sc.nextInt();

        int[] arr = new int[size];


        System.out.println("Enter the values of array");

        for(int i=0;i<size;i++) {

            arr[i]=sc.nextInt();

        }


        System.out.println("Enter the target value");

        int target=sc.nextInt();


        boolean flag=false;

        for(int i=0;i<arr.length;i++) {

            if(arr[i]==target) {

                flag=true;
```

```
                System.out.println("The value is present at index "
+i);

            }

        }



        if(flag==false) {

            System.out.println("Element not found");

        }

    }

}
```

```
Enter the size of array
5
Enter the values of array
17 7 9 11 4
Enter the target value
4
The value is present at index 4
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Object Oriented Programming (OOP)

Question 1

Question:
Create a class to represent a student with attributes like name, roll number, and marks. Implement inheritance to create a 'GraduateStudent' class that extends the 'Student' class with additional features.

```
package Kshitiz_Gupta_java_training.session1.oops;



public class Inheritance {

    public static void main(String[] args) {

        GraduateStudent s = new
GraduateStudent("Kshitiz",92,101,2026,'A',false);

        s.displayDetails();

    }





}



class Student {

    String name;

    int marks;

    int rollno;



    Student(String name,int marks,int rollno) {

        this.name=name;

        this.marks=marks;
```

```
        this.rollno=rollno;

    }

}


class GraduateStudent extends Student {

    int yearOfPassing;

    char finalGrades;

    boolean duesClear;


    GraduateStudent(String name, int marks,int rollno,int
yearOfPassing,char finalGrades,boolean duesClear) {

        super(name,marks,rollno);

        this.yearOfPassing=yearOfPassing;

        this.finalGrades=finalGrades;

        this.duesClear=duesClear;

    }


    void displayDetails() {

        System.out.println("Name: " + name);

        System.out.println("Roll No: " + rollno);

        System.out.println("Marks: " + marks);

        System.out.println("Year of Passing: " + yearOfPassing);

        System.out.println("Final Grade: " + finalGrades);

        System.out.println("Dues Cleared: " + duesClear);

    }
```

```
}
```

```
Name: Kshitiz
Roll No: 101
Marks: 92
Year of Passing: 2026
Final Grade: A
Dues Cleared: false
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 2

Question:

Demonstrate polymorphism by creating methods with the same name but different parameters in a parent and child class.

```
package Kshitiz_Gupta_java_training.session1.oops;


public class Polymorphism {

    public static void main(String[] args) {

        Child c = new Child();

        System.out.println(c.add(5,6));

        System.out.println(c.add(5,6,7));

    }

}


class Parent {

    public int add(int num1,int num2) {

        return num1+num2;

    }
```

```
}


class Child extends Parent {

    public int add(int num1,int num2,int num3) {

        return num1+num2+num3;

    }

}
```

```
11
18
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 3

Question:
Explain the concept of encapsulation with a suitable example.

```
package Kshitiz_Gupta_java_training.session1.oops;


public class Encapsulation {

    public static void main(String[] args) {

        Demo d = new Demo();

        d.setPassword(123456);

        System.out.println(d.getPassword());

    }

}
```

```
class Demo {

    private int password;


    // Getter

    public int getPassword() {

        return this.password;

    }


    // Setter

    public void setPassword(int password) {

        this.password = password;

    }

}
```

```
123456
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

String Manipulation

Question 1

Question:
Write a program to reverse a given string.

```
package Kshitiz_Gupta_java_training.session1.StringManipulation;

import java.util.*;


public class ReverseString {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the String");

        String str = sc.nextLine();


        StringBuilder sb = new StringBuilder(str);

        int start=0;

        int end=str.length()-1;


        while(start<end) {

            char temp=sb.charAt(start);

            sb.setCharAt(start,sb.charAt(end));

            sb.setCharAt(end, temp);

            start++;

            end--;

        }
```

```
        System.out.println(sb);

    }

}
```

```
Enter the String
Kshitiz Gupta
atpuG zitihsK
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 2

Question:
Implement a function to count the number of vowels in a string.

```
package Kshitiz_Gupta_java_training.session1.StringManipulation;

import java.util.*;


public class VowelCount {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the String in lowercase");

        String str = sc.nextLine();


        int counta=0;

        int counte=0;

        int counti=0;

        int countu=0;
```

```
        int counto=0;
        for(int i=0;i<str.length();i++) {
            if(str.charAt(i)=='a') {
                counta++;
            }
            else if(str.charAt(i)=='e') {
                counte++;
            }
            else if(str.charAt(i)=='i') {
                counti++;
            }
            else if(str.charAt(i)=='o') {
                counto++;
            }
            else if(str.charAt(i)=='u') {
                countu++;
            }
        }

        System.out.println( "a " + counta);
        System.out.println( "e " + counte);
        System.out.println( "i " + counti);
        System.out.println( "o " + counto);
        System.out.println( "u " + countu);
    }
```

```
}
```

```
Enter the String in lowercase
kshitiz gupta
a 1
e 0
i 2
o 0
u 1
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 3

Question:

Create a program to check if two strings are anagrams.

```
package Kshitiz_Gupta_java_training.session1.StringManipulation;

import java.util.*;


public class Anagram {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.println("Enter the First String");

        String str1=sc.nextLine();


        System.out.println("Enter the Second String");

        String str2=sc.nextLine();
```

```
        int[] freq1 = new int[26];
        for(int i=0;i<str1.length();i++) {
            freq1[str1.charAt(i)-'a']++;
        }

        int[] freq2 = new int[26];
        for(int i=0;i<str2.length();i++) {
            freq2[str2.charAt(i)-'a']++;
        }

        boolean flag=true;
        for(int i=0;i<26;i++) {
            if(freq1[i]!=freq2[i]) {
                flag=false;
                System.out.println("Not Anagrams");
                break;
            }
        }

        if(flag==true) {
            System.out.println("Strings are Anagram");
        }

    }
```

```
}
```

```
Enter the First String
kshitiz
Enter the Second String
hsiztik
Strings are Anagram
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Advanced Topics

Question 1

Question:
Explain the concept of interfaces and abstract classes with examples.

```
package Kshitiz_Gupta_java_training.session1.AdvancedTopics;


public class Abstraction {

    public static void main(String[] args) {

        Dog d = new Dog();

        d.sound();

    }

}


abstract class Animal {

    abstract void sound();


    public void eat() {

        System.out.println("Animal eats");

    }

}


class Dog extends Animal {

    void sound() {

        System.out.println("Dog barks");

    }
```

```
}
```

```
Dog barks
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

```
package Kshitiz_Gupta_java_training.session1.AdvancedTopics;


public class Oops {

    public static void main(String[] args) {

        Car c = new Car();

        c.start();

    }

}


interface Vehicle {

    void start();

}


class Car implements Vehicle {

    public void start() {

        System.out.println("Vehicle starts");

    }

}
```

```
Vehicle starts
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```

Question 2

Question:
Create a program to handle exceptions using try-catch blocks

Note : I use the same program for Question2 and 3 as it satisfies both requirements..

```
package Kshitiz_Gupta_java_training.session1.AdvancedTopics;


import java.io.*;


public class ReadData {

    public static void main(String[] args) {


        String fileName = "sample.txt";


        try {

            File file = new File(fileName);


            if (file.createNewFile()) {

                System.out.println("File created: " +
file.getName());


                FileWriter writer = new FileWriter(file);
```

```
                writer.write("Hello Kshitiz \n Welcome to Java File
Handling");

                writer.close();

            }



            BufferedReader br = new BufferedReader(new
FileReader(file));

            String line;



            System.out.println("Reading file content:");

            while ((line = br.readLine()) != null) {

                System.out.println(line);

            }



            br.close();



        } catch (IOException e) {

            System.out.println("Error: " + e.getMessage());

        }

    }

}
```

Question 3

Question:

Implement a simple file I/O operation to read data from a text file. Explore multithreading in Java to perform multiple tasks concurrently.

```
package Kshitiz_Gupta_java_training.session1.AdvancedTopics;
```

```
import java.io.*;

public class ReadData {
    public static void main(String[] args) {

        String fileName = "sample.txt";

        try {
            File file = new File(fileName);

            if (file.createNewFile()) {
                System.out.println("File created: " +
file.getName());

                FileWriter writer = new FileWriter(file);
                writer.write("Hello Kshitiz \n Welcome to Java File
Handling");
                writer.close();
            }

            BufferedReader br = new BufferedReader(new
FileReader(file));
            String line;

            System.out.println("Reading file content:");
```

```
            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();

        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

```
Reading file content:
Hello Kshitiz
 Welcome to Java File Handling
PS C:\Users\Kshitiz Gupta\Desktop\nucleusteq_assignments>
```